

Topic 4: Operators in C++

3.1 Arithmetic Operators

- `+` addition: `5 + 3 = 8`
- `-` subtraction: `10 - 4 = 6`
- `*` multiplication: `3 * 4 = 12`
- `/` division: `7 / 2 = 3` (integer division — decimal is dropped), `7.0 / 2 = 3.5`
- `%` modulus (remainder): `7 % 2 = 1`
- Integer division vs float division is a very common source of bugs — always check your types

3.2 Relational Operators

- Return true (1) or false (0)
- `==` equal to, `!=` not equal, `>` greater than, `<` less than, `>=` greater or equal, `<=` less or equal
- `5 == 5` → true, `5 == 6` → false
- Do NOT confuse `==` (comparison) with `=` (assignment)

3.3 Logical Operators

- `&&` AND: both conditions must be true — `age >= 18 && hasID == true`
- `||` OR: at least one condition true — `score >= 90 || extra_credit == true`
- `!` NOT: reverses the boolean — `!false` → `true`
- Short-circuit evaluation: `&&` stops at first false, `||` stops at first true

3.4 Assignment and Shorthand Operators

- `=` assigns value: `x = 10;`
- `+=` add and assign: `x += 5;` is `x = x + 5;`
- `-=`, `*=`, `/=`, `%=` work the same way
- `++x` prefix increment: increment first, then use the value
- `x++` postfix increment: use the value first, then increment
- `int a = 5; int b = a++;` → `b = 5`, `a = 6`
- `int a = 5; int b = ++a;` → `b = 6`, `a = 6`

3.5 Operator Precedence

- Order (high to low): `()` → `++ --` → `* / %` → `+ -` → `< > <= >=` → `== !=` → `&&` → `||`
 - `2 + 3 * 4 = 14` because `*` has higher precedence than `+`
 - Use parentheses to make intent explicit: `(2 + 3) * 4 = 20`
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